Machine Learning for DNA Artifact Classification
Low Template DNA (LTDNA) Analysis

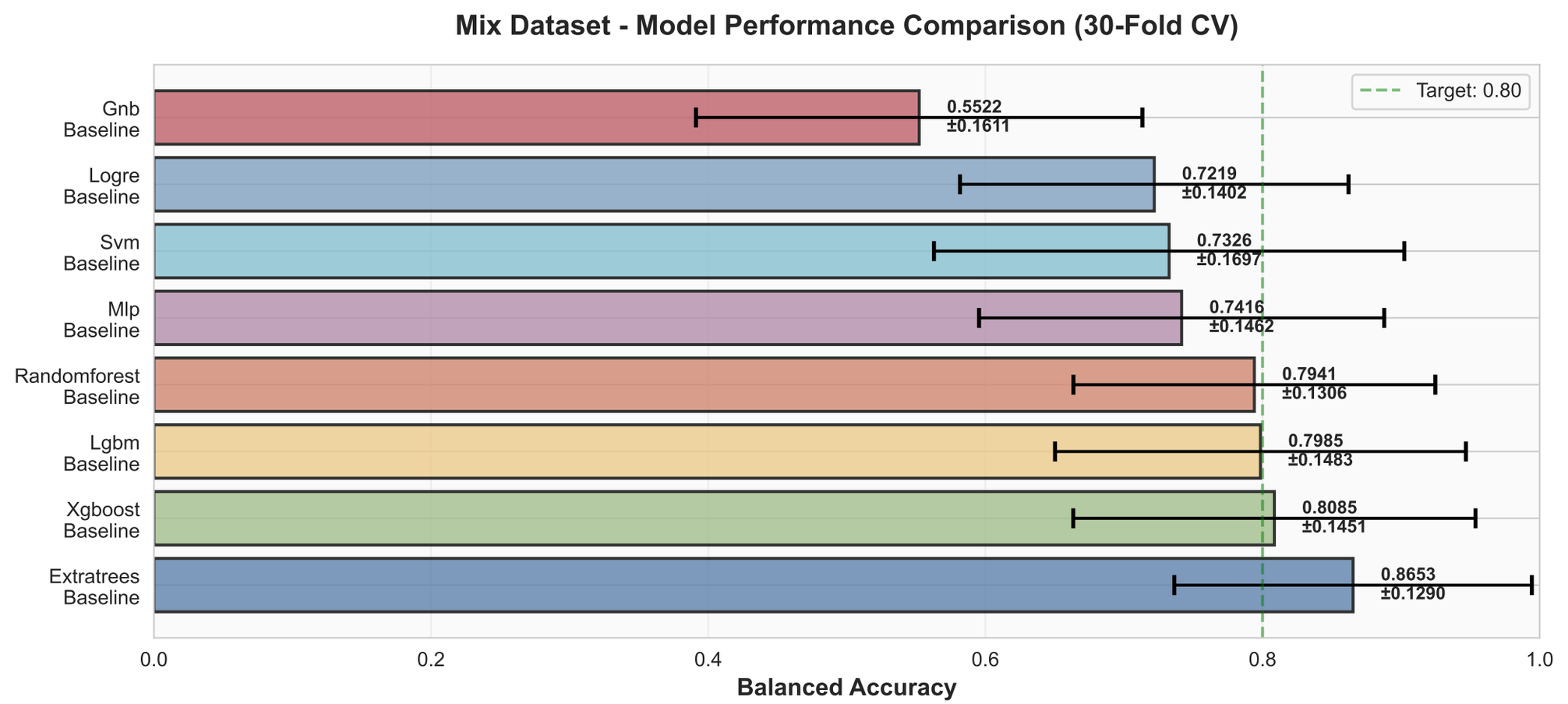
EXECUTIVE SUMMARY REPORT
Cross-Validation Results (30-Fold)

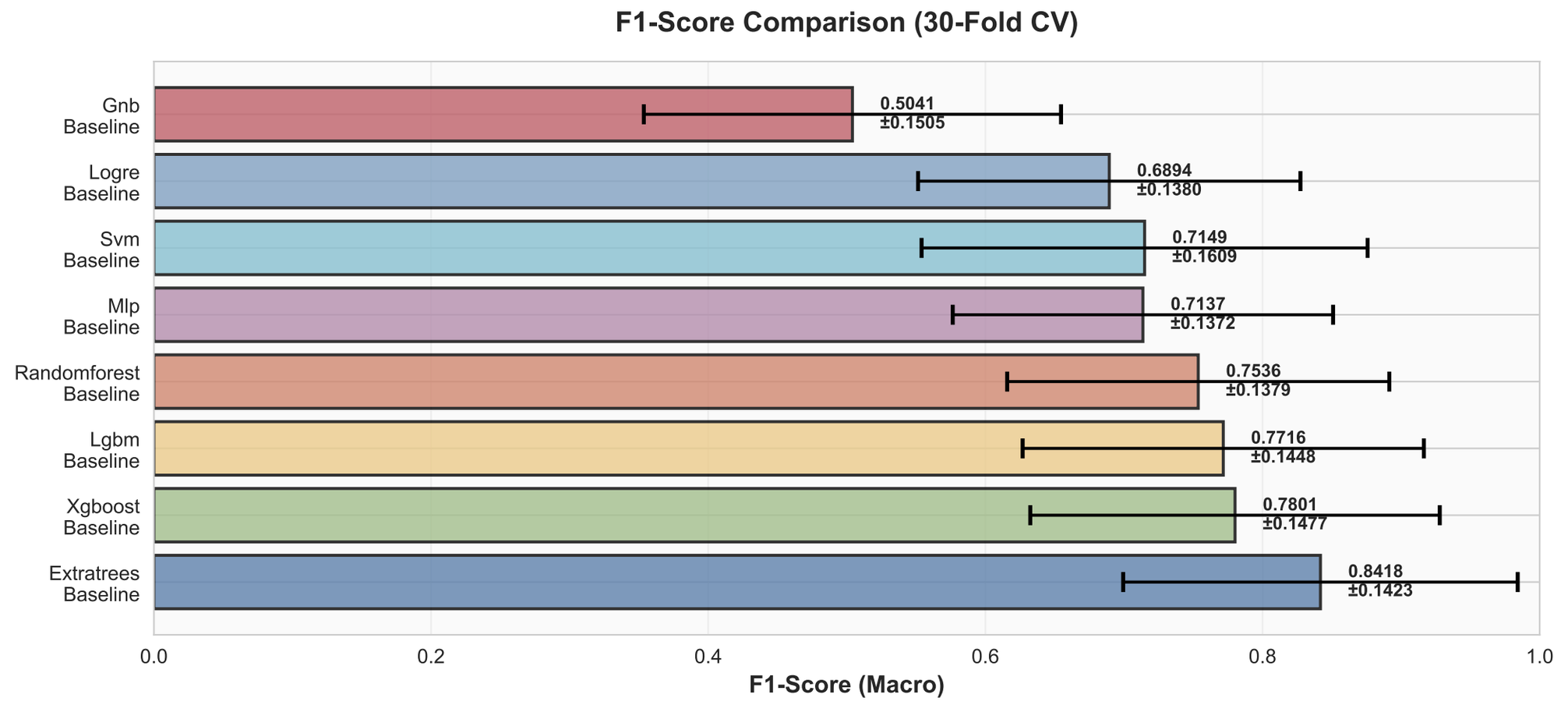
Dataset: Mix Profiles
Date: December 24, 2025
Validation: 30-Fold Stratified Cross-Validation
Models Evaluated: 8

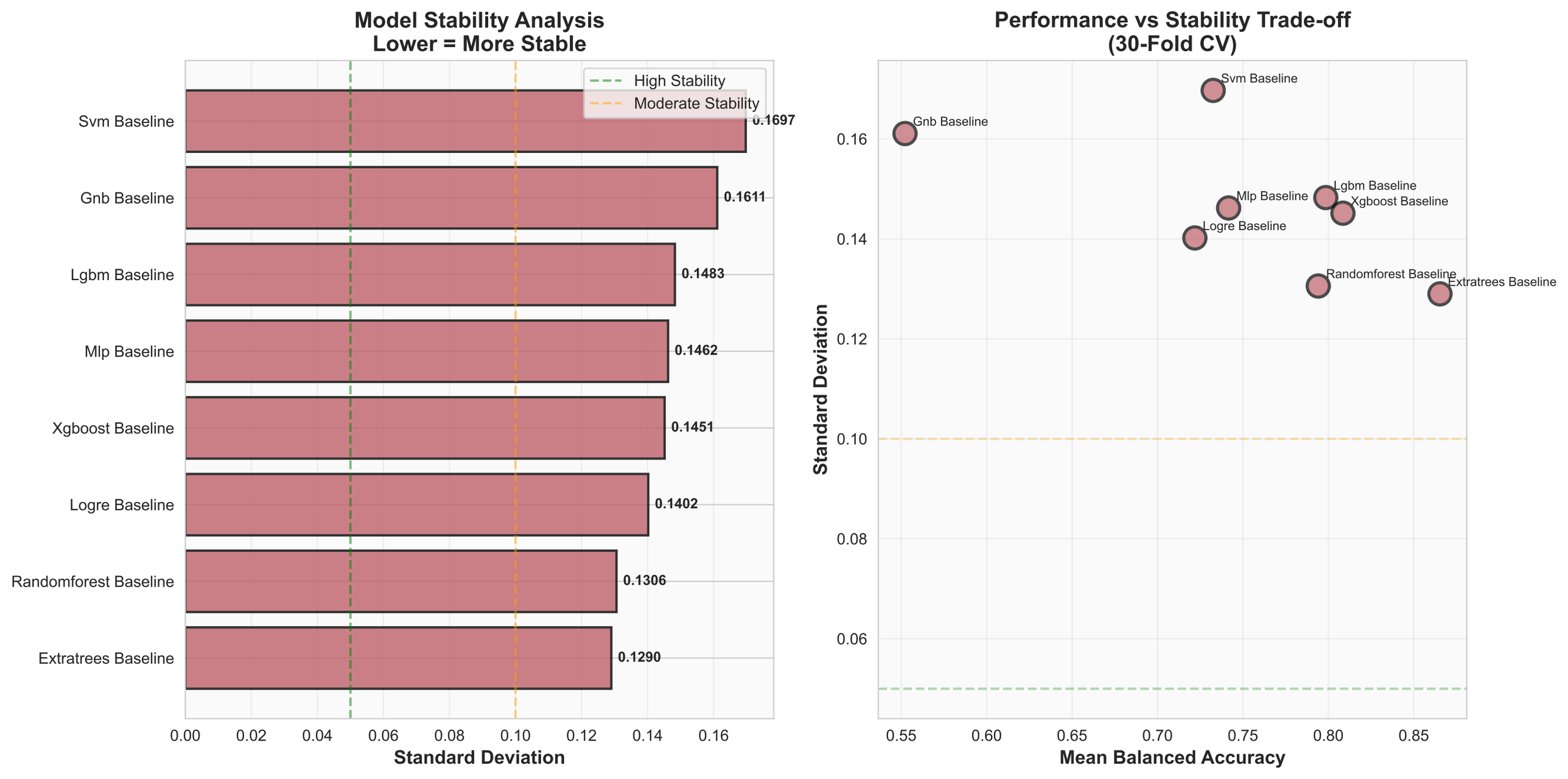
KEY FINDINGS

 Best Model: Extratrees Baseline
• Balanced Accuracy: 0.8653 ± 0.1290
• F1-Score (Macro): 0.8418
• Stability: Variable

 Validation Approach:
• 30-fold stratified cross-validation
• Ensures robust performance estimates
• Accounts for data variability
• Reduces overfitting bias


Mix Dataset - Model Performance Comparison (30-Fold CV)
Target: 0.80
Gnb Baseline
0.5522 ±0.1611
Logre Baseline
0.7219 ±0.1402
Svm Baseline
0.7326 ±0.1697
Mlp Baseline
0.7416 ±0.1462
Randomforest Baseline
0.7941 ±0.1306
Lgbm Baseline
0.7985 ±0.1483
Xgboost Baseline
0.8085 ±0.1451
Extratrees Baseline
0.8653 ±0.1290
0.0
0.2
0.4
0.6
0.8
1.0
Balanced Accuracy


F1-Score Comparison (30-Fold CV)
Gnb Baseline
0.5041 ±0.1505
Logre Baseline
0.6894 ±0.1380
Svm Baseline
0.7149 ±0.1609
Mlp Baseline
0.7137 ±0.1372
Randomforest Baseline
0.7536 ±0.1379
Lgbm Baseline
0.7716 ±0.1448
Xgboost Baseline
0.7801 ±0.1477
Extratrees Baseline
0.8418 ±0.1423
0.0
0.2
0.4
0.6
0.8
1.0
F1-Score (Macro)


Model Stability Analysis
Lower = More Stable
High Stability
Moderate Stability
Svm Baseline
Gnb Baseline
Lgbm Baseline
Mlp Baseline
Xgboost Baseline
Logre Baseline
Randomforest Baseline
Extratrees Baseline
0.1697
0.1611
0.1483
0.1462
0.1451
0.1402
0.1306
0.1290
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Standard Deviation
Performance vs Stability Trade-off
(30-Fold CV)
Svm Baseline
Gnb Baseline
Lgbm Baseline
Mlp Baseline
Xgboost Baseline
Logre Baseline
Randomforest Baseline
Extratrees Baseline
0.16
0.14
0.12
0.10
0.08
0.06
Standard Deviation
0.55
0.60
0.65
0.70
0.75
0.80
0.85
Mean Balanced Accuracy

Mix Dataset - Detailed Performance Metrics

Rank	Model	Bal Acc (Mean±Std)	F1 (Mean±Std)	Accuracy (Mean±Std)	Stability
1	Extratrees Baseline	0.8653±0.1290	0.8418±0.1423	0.8879±0.0897	Variable
2	Xgboost Baseline	0.8085±0.1451	0.7801±0.1477	0.8413±0.0970	Variable
3	Lgbm Baseline	0.7985±0.1483	0.7716±0.1448	0.8416±0.0911	Variable
4	Randomforest Baseline	0.7941±0.1306	0.7536±0.1379	0.8081±0.1017	Variable
5	Mlp Baseline	0.7416±0.1462	0.7137±0.1372	0.7886±0.0898	Variable
6	Svm Baseline	0.7326±0.1697	0.7149±0.1609	0.7870±0.1032	Variable
7	Logre Baseline	0.7219±0.1402	0.6894±0.1380	0.7792±0.0879	Variable
8	Gnb Baseline	0.5522±0.1611	0.5041±0.1505	0.5400±0.1145	Variable